

IMPACT ANALYSIS

CR033: Last Login Tracking and Display for Student Accounts

1. Change Summary

CR033 introduces last login tracking and display for student accounts in the administrator's student list. The system will record the date and time of a student's most recent successful login and display that value in a new Last Login column on the administrator dashboard.

2. Impact Assessment

Area	Impact Assessment
Frontend	Low - A Last Login column and date/time display will be added to the administrator student table.
Backend	Moderate - The authentication flow will update the last login timestamp after successful password verification, and student retrieval will return the value.
API	Low - The existing student API response will be extended to include last_login; no new endpoint is required.
Database	Moderate - A nullable last_login DATETIME column will be added to the users table.
Authentication	Low - Existing login validation remains unchanged; the timestamp is updated only after successful authentication.
Authorization	None - Existing authentication and role-based authorization remain unchanged.
Performance	Low - One timestamp update occurs after a successful login, and the existing student list query retrieves the value.
CRUD	Low - Existing Add, Edit, and Delete operations remain unchanged and must continue to work correctly.
Deployment	Moderate - A database schema migration is required before deploying the updated backend/frontend.

3. Frontend Impact

The main affected area is the administrator dashboard.

Potential changes include:

- Last Login column in the student table.
- Display of the most recent successful login date and time.
- Display of “Never logged in” when no successful login has been recorded.
- Date/time formatting for readable administrator viewing.
- Preservation of existing Edit and Delete actions.

The implementation must preserve the relationship between each displayed student and its original student_id so that existing CRUD operations continue to affect the correct student.

4. Backend, API and Database Impact

Backend changes are required to record and retrieve the last login timestamp.

The authentication flow will:

- Verify the submitted username and password using the existing authentication process.
- Update users.last_login with the current date and time only after successful password verification.
- Continue issuing the existing JWT token and user information.

The existing GET /api/students/ endpoint will continue to provide the student list, with the response extended to include the last_login value obtained by joining the students and users tables.

A database schema change is required to add a nullable last_login DATETIME column to the users table. No new API endpoint is required.

5. Security Impact

The existing authentication and authorization mechanisms remain unchanged.

The last login feature must:

- Update last_login only after successful password verification.
- Not update the timestamp when an incorrect password is submitted.
- Not bypass existing authentication or role-based authorization.
- Not expose passwords, password hashes, JWT secrets, or other sensitive authentication data.
- Preserve existing backend security controls.

6. CRUD and Functional Impact

Existing administrator CRUD operations must continue to work after the Last Login column is added.

Particular attention is required for:

- Displaying the correct last_login value for each student.
- Displaying “Never logged in” for students without a recorded successful login.
- Editing a student without affecting the last login value incorrectly.
- Deleting a student without breaking the student list.
- Maintaining the correct student_id for Edit and Delete operations.

The Last Login feature must provide additional account activity information without changing the existing student identity or CRUD behaviour.

7. Performance Impact

The expected performance impact is low. A successful login performs one additional database update to record the current login time, while the existing student list query retrieves the timestamp through the users table.

No additional request is generated from the frontend simply to display Last Login. The existing student list response provides the required value.

This approach is suitable for the current system and expected dataset size.

8. Testing Impact

Manual testing should cover:

- Successful student login updates last_login.
- Incorrect password does not update last_login.
- Student list returns the last_login value.
- Correct date/time is displayed in the Last Login column.
- Students with no previous successful login display “Never logged in”.
- Multiple successful logins update the timestamp to the latest successful login.
- Edit student functionality continues to work.
- Delete student functionality continues to work.
- Existing Add Student functionality continues to work.
- Administrator authentication and authorization remain unchanged.
- Student login and dashboard behaviour remain functional after the database change.

9. Risk Analysis

Risk	Impact / Mitigation
Last login updated after failed login	High - Update last_login only after successful password verification.
Incorrect timestamp displayed	Medium - Retrieve last_login from the related users record and format it consistently.
Existing CRUD functionality breaks	High - Perform regression testing for Add, Edit, and Delete operations.
Database migration causes deployment issue	Medium - Apply and verify the last_login schema migration before testing the application.
Student has no previous login	Low - Allow NULL and display "Never logged in".
Authentication behaviour changes unexpectedly	High - Preserve the existing password verification and JWT generation flow.

10. Overall Impact and Recommendation

Overall Impact: Moderate

CR-006 affects the database, backend authentication flow, existing student API response, and administrator dashboard. The change does not require a new authentication mechanism or a new API endpoint, and existing authorization controls remain unchanged.

The change is technically feasible and recommended for implementation, subject to applying the database migration and completing the required functional, security, and regression testing.

Key implementation consideration: last_login must be updated only after a successful login, and the displayed timestamp must correspond to the correct student account without affecting existing CRUD operations.